MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) THIRD SEMESTER PRACTICAL EXAMINATION, OCTOBER 2025

MG3DSCPHY200 – Principles of Mechanics

Duration:02hrs **Maximum Marks: 35**

Record:5 marks

Attempt the question marked 'X'. Change of question is not allowed (30 marks)

1. Determine the length of a simple pendulum that is dynamically equivalent to a given symmetric compound pendulum by measuring the periods of oscillation with its standard deviation. Comment about the observed time periods of the two pendulums . [U, A, An][8]

2. Determine the moment of inertia of a bar using it as a compound pendulum. Estimate the standard deviation in your result. [U, A, An][8]

3. Determine the moment of inertia of a flywheel. Estimate the standard deviation in your result. [U, A, An][8]

4. Determine the length of a simple pendulum that is dynamically equivalent to a given Kater's pendulum by measuring the periods of oscillation with its standard deviation. Comment about the observed time periods of the two pendulums. [U, A, An][8]

5. Determine moment of inertia of a disc using a torsion pendulum. Estimate the standard deviation in your result. [U, A, An][8]

6. Determine the Spring constant of the given spring and acceleration due to gravity by studying the oscillatory motion of spring.Estimate the standard deviation in your result. [U, A, An][8]

7. Determine the length of a simple pendulum that is dynamically equivalent to a given Asymmetric compound pendulum by measuring the periods of oscillation with its standard deviation. Comment about the observed time periods of the two pendulums [U, A, An][8]

8. Develop a Python program for solving and visualizing the dynamics of a harmonic oscillator. [U, A, An][8]

9. Develop a Python program for solving and visualizing the dynamics of a damped harmonic oscillator under different damping conditions. [U, A, An][8]

10. Compute and plot the motion of a particle under the action of the central force $F=-K/r^3(1-\alpha/r)r$, where α and K are constants. Show that this orbit precesses. Show how your choice of α and K affects the motion. [U, A, An][8]